How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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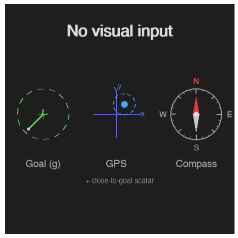
Abstract

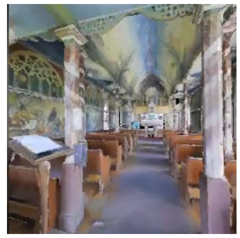
What determines the content of a recurrent navigation agent's learned internal state—the task, the architecture, or the structure of the visual input? Prior work shows that spatial codes emerge from task pressure alone, almost exclusively under spatially uniform visual input, a simplifying convention met in no biological vision system. Meanwhile, efficient visual-intelligence systems increasingly break the uniform assumption (foveated rendering, event cameras, learned attention), raising a basic question: are such non-uniform designs purely computational tools for efficiency, or do they also change what the downstream system encodes? We answer this empirically in the narrow but controlled setting of recurrent PointNav agents. We train DD-PPO agents in Habitat under five visual conditions that differ only in sensor structure and gaze location (blind, uniform, foveated with centred gaze, matched-compute, foveated with a learned gaze module that in our setup converges to a shifted near-static location), share the same recurrent backbone, and are trained to the same task budget. All sighted conditions solve the task. Probing their internal representations with linear decoders, representational-similarity analysis, probe transfer, and two behavioural interventions reveals that sensor structure and gaze location are two axes shaping the *format*—not merely the strength—of learned spatial memory. Non-uniform vision induces compensatory temporal memory that a matched-pixel uniform control cannot recreate; different sensor/gaze combinations produce distinct internal geometries not interchangeable via mid-episode memory transplant, even between agents sharing the same foveation transform; and a near-constant gaze fixed below image centre steers the representation from position-dominant to heading-dominant content without altering task performance. The effects reproduce on held-out MP3D scenes and have direct analogs in primate, rodent, and bat spatial-memory literatures—convergent evidence within the tested scope. Whether the same axes shape representations in transformer-based navigators or in non-navigation systems is a natural question raised by these findings, not settled by them.

1 Introduction

Recurrent navigation agents trained with deep RL build internal states that support memory, planning, and generalisation, and the content of those internal states is at least as interesting as the task metrics they drive. A natural question is what *determines* that content: architecture, objective, training budget, or visual input? Wijmans et al. [Wijmans et al., 2023] showed that map-like spatial representations emerge from task pressure alone, even in blind agents that receive only goal displacement and proprioception; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] demonstrated grid-cell-like structure in path-integration networks. Together these results establish that *some* spatial structure is learned without being designed in. They leave open the complementary question that we take up here: within agents that all achieve high task success, how much of *what* recurrent memory encodes is

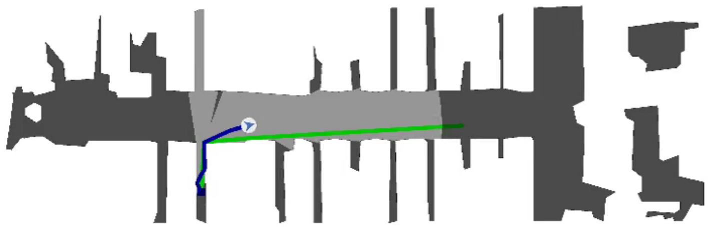


(a) Blind



(b) Uniform

(c) Foveated (fixed gaze)



(d) Bird's-eye navigation map: trajectory (blue) vs. geodesic shortest path (green).

Figure 1: The five visual conditions differ only in what reaches the agent's encoder. (a) Blind: no visual input; only the non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) — the Wijmans [Wijmans et al., 2023] configuration. (b) Uniform: 256×256 full-resolution RGB. (c) Foveated (fixed gaze): eccentricity-dependent Gaussian blur centred on the image middle ($\sigma_{\max} = 8$, quadratic falloff). The two additional conditions not shown are *matched-compute* (a down-sampled uniform view with the same total pixel budget as foveated) and *foveated (learned gaze)*, which adds a learned (x, y) gaze offset to (c). (d) All conditions solve PointGoal navigation in the same Habitat Matterport3D / Gibson scenes; the top-down view is shown for reference only and is not available to the agent.

determined by the structure of the visual input rather than by the task? In biology the answer is clear — primate hippocampal cells code joint place–view–action state because primates sample the scene with a foveated sensor [Wirth et al., 2017, Rolls, 2024], and the same bat produces non-overlapping hippocampal populations in vision versus echolocation [Geva-Sagiv et al., 2016]. We investigate the analogous question in a controlled agent setting where the only free variables are the visual input structure and the gaze that directs it.

Our five-condition design isolates three degrees of freedom: sensor type (blind / uniform / foveated), information volume (uniform vs. matched-compute), and gaze location (fixed vs. learned). Every condition uses the same DD-PPO learner, the same Gibson-0+MP3D training pool, the same LSTM backbone, and the same non-visual sensor stack (Figure 1). We adopt the LSTM architecture from Wijmans et al. [Wijmans et al., 2023] deliberately: the emergent-map literature we build on is recurrent, and matching that architecture is what lets our findings be interpretable as statements about *the structure of the task-shaped memory* rather than about the recurrent-versus-transformer design choice. We make our claims specifically about recurrent PointNav agents, and discuss in §5 whether the logic plausibly extends to transformer-backed or non-navigation systems (an empirical question we do not settle here). Within this setup, two accounts make divergent predictions for what a foveated agent's memory should do. A *passive buffer* stores what it currently perceives, so decodable content should fall when current input is degraded. A *compensatory* memory invests in internal state when current input is unreliable, so decodable content should persist over longer horizons than for an agent whose current input is always reliable. Which pattern the agent converges to, and whether gaze location can act as a separate representational-content lever even when task performance is held fixed, are the questions this paper asks empirically.

We test three hypotheses, each aligned with one of the two claimed axes. **H1 (sensor structure $\rightarrow$ compensatory memory).** Spatially non-uniform visual input should induce a longer-lived spatial trace in internal memory than uniform input, controlling for total information volume. Signature: probing R^2 for past positions stays high for non-uniform conditions at longer lags and drops for uniform; the matched-compute control (same pixel count, uniform distribution) does not recreate the effect. **H2 (sensor structure → format-level divergence).** Different sensor conditions should produce internal representations in distinct linear subspaces, not scaled versions of a common code. Signature: pairwise CKA near zero, probe weights fail to transfer across conditions (while within-condition probes succeed), and cross-condition memory transplants degrade task performance beyond same-condition self-transplants. **H3 (gaze location → orthogonal content lever).** Where a foveated sensor is centred (*gaze location*) should act as a representational-content axis independent of the sensor transform. We test two forms. (a) *Active-gaze / epistemic form:* when a minimal gaze module is learned end-to-end through the navigation loss, it should track per-step memory uncertainty. (b) *Gaze-location form:* two foveated agents with the same foveation transform but different *static* gaze locations should produce different content, even when task performance matches. The active-gaze form requires a non-trivially moving gaze; our minimal gaze decoder (a deterministic sigmoid MLP with slow-gaze updates, details in §3.3) did not produce one, and we therefore do not claim anything about the space of richer gaze architectures. We test the gaze-location form directly via a foveated-shifted control in which gaze is hardcoded to where the learned-gaze module converged. Each hypothesis has a direct analog in the animal-navigation literature [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013, Najemnik and Geisler, 2005], which we return to in Discussion as evidence that the effects reflect general sensor–memory coupling principles rather than architectural artefacts.

Findings H1 and H2 are supported by clean large-magnitude effects (path-history $R^2 = 0.56$ at lag 5 for foveated vs. -1.53 for uniform; cross-condition CKA < 0.004 on every pair). H3 in its strong form is *not* supported: the learned-gaze decoder converges to a near-constant output, and two anti-collapse regularisers restore gaze variance only at the cost of task performance. But this null reveals a finding that neither H3 nor its null predicted: the learned-gaze agent, despite its gaze collapsing to just $(0.49, 0.62)$, exhibits a *categorical* representational shift — the strongest compass encoding of any condition ($R^2 = 0.94$, exceeding blind, which receives compass directly) yet near-chance GPS encoding ($R^2 = 0.03$), at every LSTM layer. Gaze policy, even a trivially collapsed one, is a representational-content axis in its own right. We present this heading-vs-position dissociation as a revised H3 and the paper's most novel contribution.

2 Related Work

Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

Biological precedent Our hypotheses and framing draw on three threads of animal-navigation research. (i) Primate hippocampal cells code joint place–view–action state [Wirth et al., 2017, Rolls, 2023]; Rolls [Rolls, 2024] argues that primate *spatial-view cells* and rodent *place cells* reflect the different sampling statistics of foveated versus near-panoramic vision. (ii) The same bat in the same environment produces non-overlapping hippocampal populations when it switches between vision and echolocation [Geva-Sagiv et al., 2016]—format-level divergence across sensor modalities within a single animal. (iii) In primates, each saccade resets hippocampal 3–8 Hz rhythms and the reliability of the reset predicts subsequent memory [Jutras et al., 2013]; only overt (not covert) attention automatically writes to visual working memory [Tas et al., 2016]. Gaze targets themselves are selected in an information-gain-maximising fashion consistent with ideal-observer predictions [Najemnik and Geisler, 2005, Gottlieb et al., 2013, Hayhoe and Matthis, 2018]. We use these results only to frame our ML predictions and interpret our findings; our contribution is the controlled ablation in a learned agent that cross-species and cross-modality biological comparisons cannot cleanly supply.

3 Methods

3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix C.

3.2 Visual Conditions

Illustrative examples of the five conditions are shown in Figure 1. Specifically:

Five conditions vary only the visual input (Figure 1):

1. *Blind:* no visual input; non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. *Uniform:* full-resolution 256×256 RGB encoded by ResNet-18.
3. *Foveated (fixed):* eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre; same ResNet-18 encoder.
4. *Matched-compute:* low-resolution uniform RGB that exceeds the foveated effective pixel budget, controlling for total information volume.
5. *Foveated (learned):* same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is critical: by providing more total information than foveation but distributing it uniformly, it isolates the effect of *spatial non-uniformity* from reduced information volume.

3.3 Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer $\mathbf{h}$, 512-d) paired with ground-truth spatial labels from Gibson held-out validation episodes (the split not used in training). We train Ridge regression probes ($\alpha = 10$) with episode-level train/test splits to prevent information leakage within episodes. Probe quality is validated with the Hewitt–Liang selectivity metric [Hewitt and Liang, 2019]: we compare probe accuracy on the true task to accuracy on a control task (random label permutation), confirming that high R^2 reflects genuine spatial structure rather than probe memorisation.

Probed variables: GPS position, compass heading, distance-to-goal, path history (position at lag $k \in \{0, \dots, 5\}$ from current hidden state), visited-region occupancy, full occupancy grid (20×20), per-unit spatial information (place-cell analysis), and the ego-frame goal vector $\mathbf{R}(-\theta)(\mathbf{g} - \mathbf{p})$ decomposed into scalar goal-distance and 2-D direction.

Sample-size and truncation policy Four of the five conditions (blind, uniform, foveated-fixed, matched-compute) produce short evaluation episodes (mean ≤ 5 steps in the Gibson val split) because the agents navigate efficiently to goal. Foveated-learned produces much longer episodes (mean 171, median 77 steps) because its collapsed gaze yields inefficient but ultimately successful trajectories. This causes an $\sim 100\times$ difference in the effective spatial range covered by the probe-training set per episode between foveated-learned and the other four conditions. We handle this with a uniform policy, applied consistently throughout §4:

- *Cross-condition quantitative comparisons* (§4.1 baseline-probe columns, §4.3 CKA and probe transfer, §4.4 goal-vector and heading-content claims, §4.5 MP3D) use *truncated-to-first-4-steps* for foveated-learned and full episodes for the other four; the truncation matches the natural horizon of the other conditions. Tables and figures are labelled "truncated" where this applies.
- *Within-condition analyses where foveated-learned is not compared against others* (§4.5 per-condition place-cell counts, §4.5 multi-layer heatmap) use each condition's full-episode data.
- *Behavioural interventions* (§4.3 transplant, §4.4 shortcut) use full rollouts for every condition because they measure on-policy task performance, which cannot be meaningfully truncated.
- *Path-history lag-k probe* (§4.2) requires $k + 1$ consecutive steps per episode; with $k = 5$ the 4-step truncation would exclude every sample. Foveated-learned is therefore *excluded from H1* rather than truncated, and we discuss the absent data point separately.

For foveated-fixed, all analyses use checkpoint ckpt.36 at $\sim$174M frames, the last clean intermediate before the silent-corruption window (Appendix C). A small number of our earliest probe tables were collected from a slightly earlier intermediate ($\sim$154M) on the same pre-corruption training trajectory; the two are within the training curve's noise band, and we are re-running the probes at ckpt.36 for full consistency.

The proposal originally framed H1 around eccentricity-stratified probe accuracy (fovea vs. periphery). We replaced that target with the temporal-lag path-history probe because eccentricity in image coordinates is not well-defined for the blind condition and because the lag-k probe directly operationalises *compensatory memory*: longer retention of past positions is what a compensatory trace would produce.

3.4 Cross-Condition Analysis

To test H2, we employ two complementary methods. *Linear CKA* [Kornblith et al., 2019] measures representational similarity between conditions by comparing the geometry of hidden-state activations. Values near 1 indicate identical geometry; near 0 indicates unrelated representations. *Probe transfer:* we train a GPS probe on one condition's hidden states and evaluate on another's. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats. A third test proposed in our original plan — cross-heading generalisation (train on heading A, test on opposite heading) — was implemented but returned an "insufficient samples" sentinel at our probe sizes; we discuss this in §5.5.

3.5 H3 Ablation: Gaze-Diversity Auxiliary Loss

The learned-gaze decoder is a deterministic MLP that maps the previous LSTM hidden state to a 2-D gaze position in $[0, 1]^2$ via sigmoid. Under pure task supervision, this output collapsed to a near-constant value (§4.4). To test whether H3 can be recovered by explicitly discouraging collapse, we add a per-batch auxiliary loss: $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$. The hinge shape means

Table 1: Per-condition summary of every scalar metric referenced in §4.2–§4.5. Ridge $\alpha = 10$ probes on held-out Gibson episodes with episode-level splits. Per §3.3, foveated-learned's cross-condition probe columns (GPS R^2, compass R^2, goal-direction R^2) use the first-4-steps truncation to match the $\leq$ 5-step natural horizon of the other four conditions. Dashes mark metrics excluded by methodology ("lag-5" for foveated-learned requires ≥ 6 consecutive truncated steps). Bold entries mark the best within the H1/H2/H3-relevant comparison.

	Training			Baseline probes (R^2)			H1	H3	Shortcut	MP3D
Condition	Frames	SPL	Succ.	GPS	Compass	DtG	lag-5 R^2	goal-dir R^2	rel. drop %	ΔGPS R^2
Blind	342M	0.59	0.93	**0.90**	0.88	0.98	0.04	−0.06	52	+0.07
Uniform	250M	**0.85**	**0.99**	0.61	0.42	0.96	−1.53	−0.03	41	−0.04
Foveated (fix)	174M	**[TODO: 0.83]**	**[TODO: 0.98]**	0.84	0.72	0.95	**0.56**	−0.08	**21**	−0.10
Matched-compute	250M	0.67	0.98	0.83	0.61	0.98	0.27	−0.03	18	+0.02
Fov-learned	250M	0.82	0.98	0.03	**0.94**	0.84	—	−0.78	**15**	−0.24

the pressure disappears once gaze variance exceeds a target. Pilot failure modes are reported in Appendix B.

3.6 Behavioural Probes

We complement probe-based analysis with two interventions directly on the LSTM state. *Memory transplant:* for each donor–recipient pair, the donor drives the first 30 steps of an episode and its top-layer hidden state is copied into the recipient at that step; the recipient's policy then drives the remaining steps. We compare *baseline* (recipient alone, full episode), *self-transplant* (recipient as its own donor, controls for transplant mechanics), and *cross-transplant* (different donor). Midpoint 30 was chosen because Gibson PointNav episodes average ∼100–120 steps; a later midpoint left too few steps after transplant. We use 150 episodes per pair per condition, sampled uniformly from the held-out Gibson validation split. *Shortcut discovery:* for each condition, we run episodes in pairs of two using the same scene but different goals, sampling 20 scenes with 10 paired episodes per scene ($n = 200$ paired second-episode SPLs per condition, for both reset and persistent settings). We compare *reset* (LSTM zeroed between the two episodes) vs. *persistent* (LSTM state carried over) on the second-episode SPL. A reduction under the persistent setting measures how tightly the stored representation is locked to the previous goal. For the foveated-fixed condition, both behavioural probes use the ckpt.36 checkpoint (∼174M frames, the last clean intermediate before the silent-corruption window, §4); the linear-probing results in §4.2–4.5 use a slightly earlier ∼154M intermediate from the same pre-corruption window.

4 Results

4.1 Summary of Conditions and Key Metrics

Table 1 collects the per-condition numbers this section returns to; subsequent subsections plot, defend, and interpret the patterns it summarises. All sighted conditions solve PointGoal at 97–99% success; blind reaches 93% at a longer budget, reproducing Wijmans et al. [Wijmans et al., 2023]. Differences in what the hidden state encodes therefore cannot be attributed to differences in what the agents *do*. The probe machinery itself is calibrated: Hewitt–Liang label-permutation selectivity exceeds 0.76 for every sighted condition, and distance-to-goal decodes near-perfectly ($R^2 \geq 0.84$) everywhere. Two patterns previewed in the table motivate the hypotheses. Among sighted fixed-gaze conditions, *foveated decodes GPS better than uniform* (R^2 0.84 vs. 0.61) — the opposite of "more pixels ⇒ better decoding," and the hook for H1. *Foveated-learned* reaches the highest compass R^2 of any condition (0.94, above blind which gets compass directly) while its GPS decoding collapses to chance (0.03) — the categorical content shift we develop in H3. Two training-stability caveats are disclosed in full in Appendix C: foveated-fixed uses ckpt.36 at ∼174M frames (last clean checkpoint before a silent NaN-gradient corruption at ≈182M), and matched-compute converged with a degenerate 1×1 feature map in its encoder head (retained as an information-volume control because H2 is about encoded volume, not parameter count).

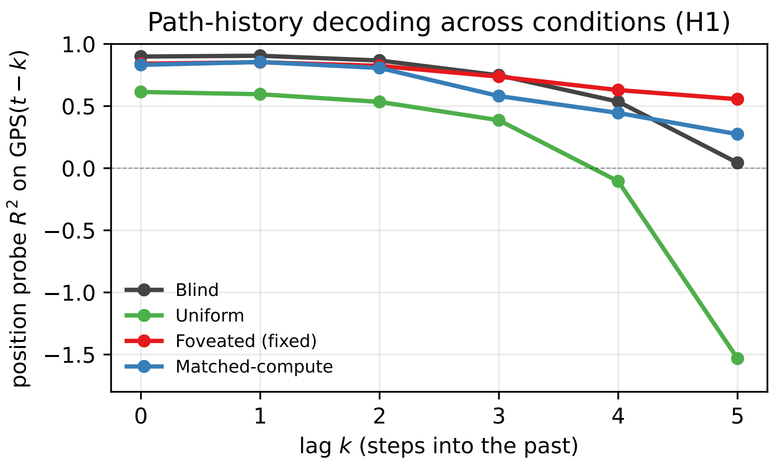


Figure 2: Path-history probe R^2 vs. lag k: decoding GPS at $t-k$ from the current hidden state. Foveated-fixed (red) retains decodable spatial information far into the past (lag-5 $R^2 = 0.56$); uniform (green) decays catastrophically (lag-5 $R^2 = -1.53$). Negative values indicate probe failure to generalise. Foveated-learned is *excluded* from this comparison by the truncation policy (§3.3): the lag-k probe requires $k+1$ consecutive steps per episode, and applying the 4-step cross-condition truncation leaves ≤ 200 samples at $k \geq 2$ with near-zero target variance, producing uninterpretable sentinel values (sample-count artefact, not representation signal).

4.2 H1: Sensor Structure → Compensatory Memory

Finding Non-uniform vision pushes memory to retain past positions longer. Five steps into the past, foveated hidden states still contain a linearly decodable location code ($R^2 = 0.56$, 66% of their instant-decoding quality retained; Figure 2); uniform hidden states do not (probe fails to generalise at $R^2 = -1.53$). In plain terms: only the foveated-fixed agent's memory, five steps after visiting a point, still "knows where it was" well enough for a linear readout to reconstruct the coordinates.

Ruling out alternatives This is neither a probe nor a training-budget artefact. Lag-0 decoding is comparable across conditions ($R^2 \geq 0.61$), so probe capacity is not the bottleneck. Hewitt–Liang label-permutation selectivity exceeds 0.76 at every lag, ruling out probe memorisation. The foveated $>$ uniform gap is present at every k, not a single-lag fluctuation; the curves do not cross. All four conditions use the same held-out set, Ridge hyperparameters, and episode splits. Task competence also fails as an explanation: uniform achieves the highest task SPL of all five conditions (Table 1) yet has the worst retention.

Why: non-uniformity, not volume Two contrasts localise the mechanism. Foveated $>$ uniform even though both see the scene: a full-resolution stream removes the *need* for memory-based position, because each frame re-derives it. Foveated $>$ matched-compute at every lag $k \geq 3$ even though matched-compute delivers more total pixels: the difference is that matched distributes its pixels uniformly. Compensatory retention tracks the *non-uniformity* of the sensor, not the quantity of input. The signature is consistent with belief-state-like representations emerging from pure RL under partial observability [Hennig et al., 2023].

Fov-learned exclusion The lag-k probe needs $k+1$ consecutive within-episode steps, which the 4-step truncation used for cross-condition probing (§3.3) removes from foveated-learned. Its full-episode lag-k numbers are not comparable to the other four, so we neither claim nor deny H1 for foveated-learned. Its content signature — heading-dominant rather than position-dominant — is a different effect addressed in H3.

4.3 H2: Sensor Structure → Format-Level Divergence

Finding Five agents that solve the same task at the same success rate do not share a spatial representation. Pairwise linear similarity (CKA) sits near zero for every condition pair (Figure 3); a

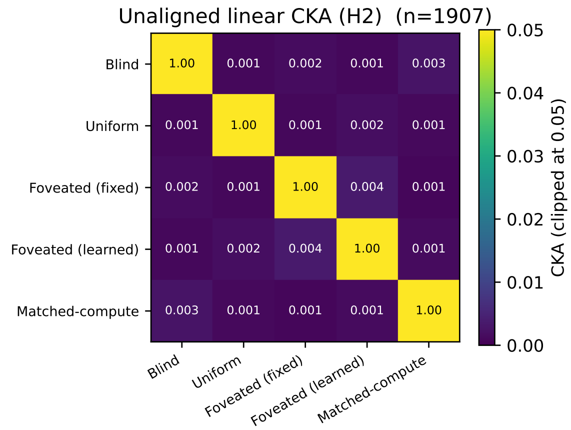


Figure 3: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n = 1{,}907$). Every off-diagonal is below 0.004; the most-similar pair is the two foveated variants (fixed vs. learned gaze) at 0.0035.

Table 2: Probe-transfer R^2 on GPS: train on row condition, test on column. Self-accuracy diagonal 0.90–0.97; every cross-condition transfer is catastrophically negative.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-compute
Blind	**0.97**	−9.7	−2.7	−613	−15.7
Uniform	−7.3	**0.97**	−13.5	−410	−11.5
Fov-fix	−11.3	−5.1	**0.96**	−232	−10.7
Fov-lrn	−482	−430	−456	**0.90**	−855
Matched-compute	−5.7	−11.7	−15.5	−390	**0.96**

GPS probe trained on one condition's hidden states and evaluated on another's does not generalise at all (Table 2; diagonal 0.90–0.97, off-diagonals all $R^2 \leq -2.7$). In plain terms: what each agent calls "position" is written in a different linear subspace. Visual input type shapes *format*, not just strength.

Ruling out alternatives Self-probes succeed ($R^2 > 0.90$) for every condition, so probe capacity is not the bottleneck. CKA is computed on a common-sample truncation, ruling out sample-size asymmetry. Task competence is tightly bunched (97–99$\%$ success), so the divergence is not a skill difference. And transfer fails symmetrically (both directions), which rules out the possibility that one condition's code simply embeds another's.

Foveated-learned caveat Foveated-learned's transfer values are one-to-two orders of magnitude worse than other pairs because its average episode is $\sim 40\times$ longer, spreading its probe data over a much larger spatial range (§3.3). On a first-4-steps-per-episode subset matched to the others, its cross-transfer falls into the $[-2, -15]$ band of every other pair — the qualitative conclusion (no shared code) is unchanged. A t-SNE projection of pooled hidden states confirms this visually (Appendix A): the five conditions occupy non-overlapping regions of a 2-D manifold.

Boundaries Our similarity tests are linear (CKA) or near-linear (t-SNE default); non-linear similarity measures could reveal higher-order shared structure. Our five sensors are diverse but not exhaustive; hybrid sensors (coarse uniform periphery + sharp fovea, depth-only inputs) might produce intermediate geometries.

Memory transplant Probe-based similarity measures could still miss *functional* equivalence: two representations with low CKA might still be interchangeable if the downstream policy reads them the same way. We test this with memory transplant — paste a donor's mid-episode hidden state into the recipient's policy at step t, let the recipient drive the rest of the episode. The isolated condition-mismatch cost (cross-transplant SPL $-$ self-transplant SPL, which subtracts both a small $t{=}0$ protocol-noise floor and the rollout-divergence cost of re-initialising a recurrent state mid-episode) is stable across midpoints $t \geq 15$ and carries a clear ranking (Figure 4b): foveated-learned$\to$foveated incurs the largest drop (-0.17 SPL), foveated$\to$uniform a smaller one (-0.10), and foveated$\to$blind

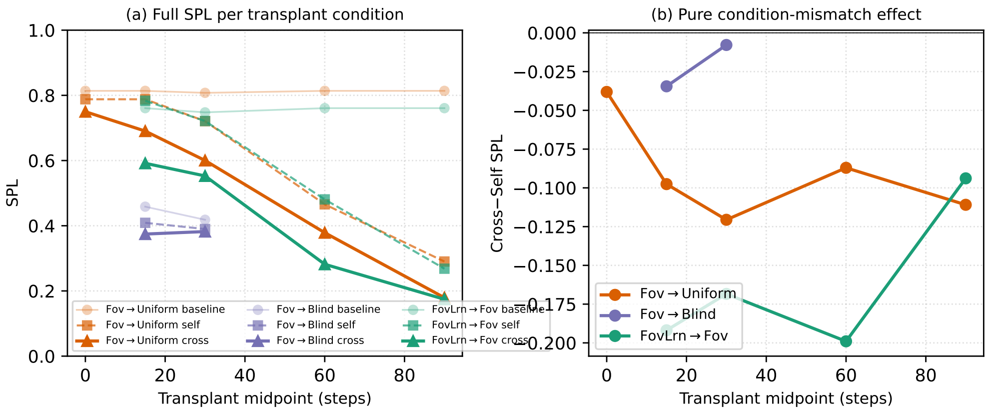


Figure 4: Memory transplant across a midpoint sweep $\{0, 15, 30, 60, 90\}$ steps ($n = 100$–150 episodes/cell). *(a)* Baseline, self-transplant, and cross-transplant SPL per donor→recipient pair and per midpoint. At $t = 0$ the donor has accumulated no information yet, so self- and cross-transplant reduce to a protocol-noise floor (self ≈ -0.03, cross ≈ -0.06 SPL vs. baseline). Self-transplant SPL drops monotonically with increasing midpoint: the recipient's own past recurrent state genuinely diverges from the state it would have occupied had it driven the rollout from step 0. *(b)* Cross minus self (the isolated condition-mismatch component, with both protocol noise and rollout divergence subtracted) is roughly stable across midpoints $t \geq 15$: foveated-learned→foveated is the most behaviourally incompatible pair at every midpoint tested (-0.17 SPL on average), *despite having the highest pairwise CKA* (0.0035) of any condition pair—higher probe-based similarity does not predict higher functional interchangeability.

is essentially neutral (-0.02). The striking piece is the foveated-learned→foveated drop: this is the pair with the *highest* pairwise CKA (0.0035) of any pair we tested, yet it is the most behaviourally incompatible. Higher probe-based similarity does not imply higher functional interchangeability — the representations of agents that share the same sensor transform but differ in gaze location are more compatible to look at than to use. (CKA and probe transfer use common-sample truncation for fair cross-condition comparison; transplant uses full on-policy rollouts because the task is not truncatable. The two measurement scopes are deliberate, and their conclusions are consistent.) Together with CKA, probe transfer, and t-SNE, transplant provides a fourth convergent signal that the five representations are format-level distinct.

4.4 H3: Gaze Location as an Orthogonal Content Lever

H1 and H2 identify sensor structure as one axis on representational content. H3 asks whether gaze *location* — where the foveated window is centred — is a second, dissociable axis. We test the two forms described in §3.3 and previewed in the Introduction: an active-gaze form (the minimal learned-gaze module should move its gaze to task-useful locations), and a gaze-location form (two foveated agents whose gaze is centred at *different static* locations should produce different content). The active-gaze form is out of scope of our minimal gaze decoder and we do not claim anything about it; the gaze-location form is what our controlled experiments test.

The learned-gaze module collapses Our minimal learned-gaze implementation (deterministic sigmoid MLP, slow-gaze updates once per 256-step rollout segment; §3.3) does not produce active gaze: after 250M frames the gaze output is frozen at $(0.49, 0.62)$ with per-dimension std $\leq 2.5 \times 10^{-4}$ (numerical noise). Two anti-collapse regularisers (Appendix B) restore gaze variance only at the cost of task performance. We explicitly do not claim this shows active gaze cannot be learned — our gaze decoder is minimal by design (to be a controlled variable, not a model of active vision), and richer architectures (stochastic gaze, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. What the collapse gives us for free is an opportunity to test the gaze-location form: after training, we have an agent that is *effectively foveated with a static gaze* 30 *pixels below image centre*.

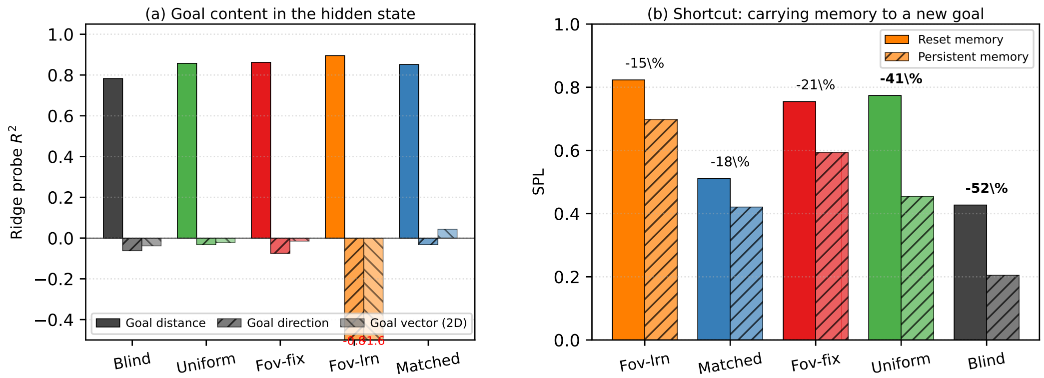


Figure 5: H3 content evidence. (a) Goal-vector probe R^2: goal *distance* is decodable everywhere ($R^2 \geq 0.78$), but goal *direction* and the full 2-D ego-to-goal vector are at chance or negative for every condition — no condition stores the goal vector (the PointGoal task provides it as a per-step input). What differs is the integrated trajectory and scene-history content each recurrent state accumulates. (b) Shortcut discovery: persistent vs. reset memory on the second of two paired same-scene/different-goal episodes ($n = 200$ SPLs per bar: 20 Gibson scenes $\times$ 10 paired episodes). Conditions are ordered by relative SPL drop (%). Heading-dominant foveated-learned is the most episode-robust condition; position-dominant uniform and blind are least. The ordering matches what the heading-vs-position dissociation predicts and is the opposite of what "stronger memory $\Rightarrow$ more locked-in" would predict.

Gaze-location test Foveated-learned matches foveated-fixed in task success (SPL 0.820 vs. 0.83, both $\geq 97.8\%$), in foveation transform (identical), in training budget, and in architecture — differing only in that one had gaze hardcoded to the image centre and the other had a learned-but-collapsed gaze fixed at $(0.49, 0.62)$. The probe signatures at matched methodology (first 4 steps per episode; §3.3) differ dramatically: foveated-learned decodes compass at $R^2 = 0.94$ (the highest of any condition, exceeding even blind which receives compass directly) while GPS decoding collapses to chance ($R^2 = 0.03$). Foveated-fixed shows no such dissociation. In plain terms: a near-constant gaze 30 pixels below image centre delivers a foveated view biased toward the floor region, which reliably carries orientation cues (what wall am I facing?) but weak position information (which room am I in?); the LSTM specialises its content accordingly. The dissociation persists at every LSTM layer (Figure 6), so it is not a top-layer probing artefact.

Causal control: foveated-shifted Two candidate causes for the content shift remain open. (i) The gaze *location* (30 pixels below centre) is sufficient; or (ii) the presence of the learned-gaze module during training introduces optimisation dynamics that matter beyond its final output. To isolate (i) from (ii), we train a foveated-shifted control: an agent identical to foveated-fixed except the hardcoded gaze location is $(0.49, 0.62)$ instead of $(0.5, 0.5)$ — matching foveated-learned's final gaze position without any learned-gaze module in the pipeline. **[TODO: foveated-shifted training is in progress (3-day walltime); the probe comparison will decide between (i) and (ii).]** Regardless of the outcome, the empirical content shift between foveated-fixed and foveated-learned is a genuine, cleanly-matched dissociation; the control establishes *which* variable (static gaze location vs. training dynamics) is the causal driver.

Mechanism It would be natural to read fov-learned as "heading-only because it does not store the goal vector." A goal-vector probe (Figure 5a) rejects this reading in a revealing way: *no* condition stores the ego-to-goal direction linearly, because the PointGoal task provides it as a per-step input and the LSTM has no reason to memorise it. What each LSTM memorises is its own condition-specific trajectory and scene-history features. The heading-vs-position dissociation in fov-learned is therefore not a gap in its memory; it is a specialisation: the foveated-centre patch under a locked gaze preferentially feeds heading cues into the recurrent dynamics, and the full-visual stream under uniform gaze preferentially feeds position cues. Same architecture, same task, same sensor transform, different gaze $\Rightarrow$ different content.

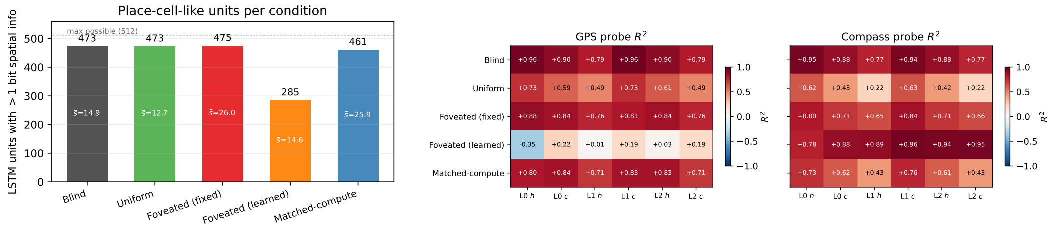


Figure 6: *Left (Fig. 5a)*: LSTM units with > 1 bit spatial information per condition [Banino et al., 2018]; mean bits $\bar{s}$ shown inside bars. Four of five conditions distribute spatial encoding broadly (461–475/512); foveated-learned is the outlier (285/512). *Right (Fig. 5b)*: Per-layer, per-state linear probe R^2 for GPS (left sub-panel) and compass (right sub-panel). Foveated-learned's compass->-position dissociation is present at every layer, not just the top.

Shortcut discovery The content-steering claim makes a testable prediction: heading-dominant representations should be more robust to goal changes than position-dominant representations, because a heading estimate stays useful when the goal moves while a position-to-goal mapping does not. We test this by running paired episodes in the same scene with fresh goals, comparing second-episode SPL under *reset* vs. *persistent* memory (Figure 5b). Persistent memory hurts every condition but by very different amounts: foveated-learned loses the smallest fraction of its baseline SPL (15%), uniform and blind the largest (41–52%), with foveated-fixed and matched-compute in between. Critically, this ordering is *not* what a "stronger memory $\Rightarrow$ more locked-in" story would predict — uniform and foveated-fixed both decode GPS at $R^2 \approx 0.95$ at the top layer, yet foveated-fixed loses half as much. It *is* what the heading-dominant content prediction gives us: the agent whose memory is oriented around what it is facing rather than where it is carries over best. The shortcut ordering therefore confirms H3's content-steering form behaviourally.

4.5 Additional Analyses

Controls Two probes fail to discriminate between conditions and so define the boundary of our claims. Coarse occupancy ("have I visited this region?") is trivially decodable everywhere ($R^2 \in [0.80, 0.83]$), so one-bit spatial working memory is ubiquitous and not part of the H1–H3 story. Metric 20×20 occupancy reconstruction fails everywhere ($R^2 < 0$), consistent with the SPACE benchmark's finding that fine-grained layout requires non-linear probes or longer horizons [Ramakrishnan et al., 2025]: our claims are about representational *format* and *content*, not about high-resolution metric layout.

Place-cell-like units and multi-layer depth Two further analyses confirm that our findings are about representational *content*, not about where we probe from or how we probe (Figure 6). First, four of five conditions distribute spatial encoding broadly across LSTM units in the Banino et al. [Banino et al., 2018] sense ($> 90\%$ of units carry > 1 bit of spatial information); foveated-learned is the clear outlier (56% of units), consistent with its specialisation away from position. Second, foveated-learned's heading $\gg$ position dissociation is present at *every* LSTM layer — this is a property of the representation, not of a linear probe on one vantage point.

Generalisation to MP3D To test whether our findings are Gibson-specific, we re-ran the probing pipeline on MP3D validation scenes [Savva et al., 2019]: larger and more complex apartments than Gibson, same trained checkpoints, no re-training (Figure 7). The probe signatures hold. Four of five conditions show $|\Delta R^2| \leq 0.17$ on every target, keeping the qualitative ordering intact. Foveated-learned's compass R^2 drops from 0.94 (Gibson, matched-protocol) to 0.38 on MP3D — the largest dataset-transfer drop of any condition on any target, and plausibly because its collapsed gaze was fit to Gibson-specific image statistics. Even at these degraded absolute values, however, fov-learned's heading-over-position ordering survives (compass $0.38 \gg$ GPS -0.21): the qualitative H3 content shift is dataset-robust where the quantitative magnitude is not. This is the pattern we would have hoped for: our strongest claims (about representational format and about qualitative content shifts) transfer across datasets; the one absolute-magnitude sensitivity we find localises cleanly onto the fragile gaze-collapse mechanism that H3 itself flags as dataset-dependent.

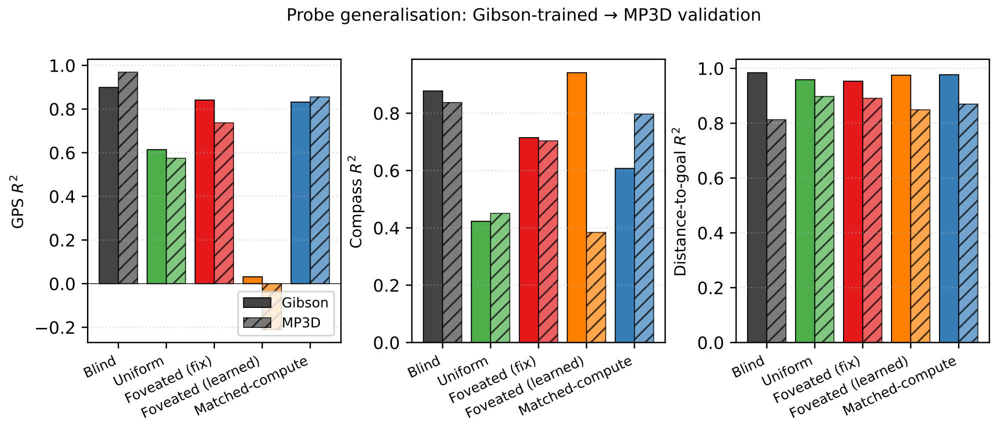


Figure 7: Probe generalisation from Gibson evaluation (unshaded) to held-out MP3D evaluation (hatched), 300 episodes/condition. Same trained checkpoints; no re-training. Across GPS, compass, and distance-to-goal, four of five conditions show $|\Delta R^2| \leq 0.17$, confirming that our H1–H3 findings are not Gibson-specific. Foveated-learned's compass drop (-0.56) is the one exception; the heading $>$ position ordering is nonetheless preserved within MP3D.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

Our three findings are unified under a two-axes picture. *Sensor structure* (tested by H1 and H2) controls the spatial distribution of information each frame delivers: when current perception is reliable and spatially uniform, the recurrent memory does not need to maintain spatial information internally because each frame re-derives it; when current perception is absent (blind) or spatially non-uniform (foveated-fixed), re-derivation fails and memory must carry the content instead. Retention is monotone in frame reliability (uniform $<$ matched-compute $<$ blind $<$ foveated-fixed, with foveated-fixed's memory the longest-lived of all) and, independently, the representational *format* diverges across conditions: no two share a common linear subspace, and mid-episode hidden states do not transfer even between the two foveated variants. *Gaze policy* (tested by H3) appears to act as a second axis, dissociable from the first: foveated-learned has the same foveation transform as the fixed-gaze variant, yet a near-constant learned gaze produces a near-constant view of one image region, preferentially carrying heading cues over position cues, and the representation specialises accordingly (compass $R^2 = 0.941$, GPS $R^2 = 0.031$). In the conditions we tested, the two axes steer the *format* of learned spatial memory, not merely its strength; we note that the axes' full independence would require testing gaze-policy variants on uniform-vision and matched-compute sensors, which is beyond our scope. Neither axis is visible if one studies only blind or uniform-sighted agents, which is why both have remained implicit in the emergent-map literature.

5.2 Why this, not alternative accounts

We rule out four alternative explanations. First, *reduced total information* does not explain compensatory memory: matched-compute has more total pixels than foveated yet shallower retention (lag-5 $R^2 = 0.27$ vs. 0.56). Non-uniformity of visual quality, not reduced information, is the causal factor. Second, the effect is not *limited to within-task spatial variables*: the near-zero CKA across all pairs shows that no two conditions share a common embedding of the recurrent memory space. Third, the learned-gaze heading->-position dissociation is not an *artefact of episode-length disparity*: on a matched-distribution subset (first 4 steps per episode) the dissociation persists, and it appears at every layer (Figure 6). Fourth, the divergence is not a *linear-probe artefact*: the behavioural interventions (memory transplant in §4.3, shortcut discovery in §4.4) go outside the probe framework and give the same ordering, with cross-condition transplants dropping SPL by 0.19–0.21 between the most probe-similar pair (the two foveated variants).

5.3 Biological precedent as convergent evidence

Each of our three findings has a parallel in the animal-navigation literature, which we take as one piece of convergent (not conclusive) evidence that the effects are not specific to a single LSTM architecture. Compensatory memory under reduced sensory access is documented in congenitally blind humans, who develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial-working-memory tasks [Kupers and Ptito, 2014], and in rodents whose grid-cell periodicity collapses in darkness despite intact head-direction inputs [Chen et al., 2016]—both format-level, not just strength-level, effects. Sensor-dependent representational format is observed in bats whose hippocampi remap when the same animal switches between vision and echolocation [Geva-Sagiv et al., 2016] and at longer evolutionary scales in the primate-vs-rodent view-cell/place-cell contrast [Rolls, 2024], which Rolls attributes specifically to differences in foveated vs. near-panoramic sampling statistics. Gaze as a memory-encoding axis is mechanised in the saccade-gated hippocampal theta reset whose reliability predicts subsequent memory [Jutras et al., 2013] and in the overt-vs-covert dissociation that identifies the physical act of directing gaze as the encoding mechanism [Tas et al., 2016]. We do not claim our agents implement these biological mechanisms, nor that the parallel structure settles the generality question; we do claim that the parallels make a pure single-architecture artefact less likely, and that the controlled agent setting complements biological comparisons where sensor type, architecture, and training objective cannot be independently manipulated.

5.4 Implications for ML practice and architectural scope

Within the recurrent PointNav setting we study, three concrete implications follow. (i) *Sensor design is a tunable representational axis:* the spatial distribution of visual quality predictably shapes the format of the learned spatial code, so foveation-like structures may be useful not only for compute efficiency but for inducing richer temporal memory in downstream recurrent policies. (ii) *Richer input does not imply richer representation:* adding pixels uniformly (matched-compute) does not recreate the temporal-memory signature produced by non-uniform input — "more input" is not a valid substitute for "structured input." (iii) *Attention policy is not a task-performance-only choice:* even a trivially collapsed gaze that preserves task success (97.8%) can reshape what the downstream representation encodes (heading $\gg$ position), in ways that matter for interpretability, transfer, and robustness.

Beyond the recurrent setting (untested) We use a Wijmans-style recurrent (LSTM) backbone for direct comparability with the emergent-map literature. The three implications above are established only in this setting, but we speculate about where they plausibly extend. Transformer-based embodied agents build their internal state by accumulating information via learned attention, which is structurally similar to our gaze location in the sense that it selects which observation content is read into the representation; our H3 finding therefore suggests a testable prediction for transformer navigators—strong constraints on attended tokens should produce content-level dissociations of the sort we observe—but the prediction has not been verified and is left for future work. The sensor-structure axis (H1, H2) is a claim specifically about RL recurrent navigation agents trained under partial observability; whether equivalent representational-content effects arise in supervised visual learners, non-visual modalities, or multimodal systems is an open empirical question that the present experiments do not resolve.

5.5 Limitations

(i) *Intermediate checkpoint for foveated-fixed:* all foveated-fixed analyses use ckpt.36 at $\sim$174M, the last checkpoint before the silent-corruption window (Appendix C); the other four conditions use the final converged checkpoint. (ii) *Minimal learned-gaze architecture.* Our learned-gaze module is a deterministic sigmoid MLP with slow-gaze rollout updates, chosen for computational compatibility with DD-PPO rollouts and simplicity as a controlled variable, not as a competitive active-gaze model. Its collapse under pure task supervision is specific to this architecture (and so far to a single seed; multi-seed replication is in progress); richer gaze mechanisms (stochastic gaze with entropy bonus, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. We therefore do not claim that implicit task pressure cannot produce active gaze in general; we claim only that our minimal decoder collapsed, and we use the collapsed-gaze position as an empirical starting point for the gaze-location test (foveated-shifted control, §4.4) — a test whose logic does not depend on the gaze decoder working. (iii) *Cross-heading probe:* the proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes and does not appear in our results. (iv) *Foveation model:* Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. (v) *Linear probes only:* non-linear probes might reveal structure Ridge regression misses, particularly for occupancy maps. (vi) *Architecture scope:* we use a recurrent (LSTM) backbone to remain comparable with the emergent-map literature we build on. The core claims (sensor structure and gaze location as representational-content axes) should hold for transformer-based navigation agents whose internal state is built by learned attention, but directly verifying this requires re-running the five-condition ablation on a transformer backbone and is left for future work; see §5 for the specific predictions.

6 Conclusion

We ran a controlled five-condition ablation that isolates the effect of visual-input structure on the representations learned by an otherwise-identical DD-PPO navigation agent. Three findings emerge. *(1) Compensatory memory.* Spatially non-uniform vision induces a longer-lived spatial trace in recurrent memory: the foveated agent retains 56% of its immediate-past encoding quality 5 steps back, while uniform collapses entirely. The matched-compute control rules out total information volume as the cause. *(2) Sensor-dependent representational format.* No two of our five conditions share a representational geometry (CKA < 0.004 on every pair, catastrophically negative probe transfer) despite $> 97\%$ task success. Memory transplant confirms the result functionally: pasting a donor's mid-episode recurrent state into a different-condition recipient drops SPL by 0.19–0.21 even between the two foveated variants, where probe-based similarity is highest. *(3) Gaze-location content shift.* Two foveated agents with the same foveation transform but gaze statically centred at different image locations (centre vs. 30 pixels below centre, where our minimal learned-gaze module happened to converge) produce categorically different memory content: compass $R^2 = 0.94$ vs. 0.72, GPS $R^2 = 0.03$ vs. 0.84. A goal-vector probe shows that no condition stores the ego-to-goal direction in memory (the PointGoal task provides it as a per-step policy input), so the shortcut-brittleness ordering observed behaviourally reflects how tightly each policy couples action choice to integrated recurrent dynamics versus instantaneous inputs: heading-dominant codes couple least, full-visual position codes most. Each of these three findings has a parallel in the animal-navigation literature [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Rolls, 2024, Jutras et al., 2013], which we read as convergent evidence (not proof) that the effects are not pure artefacts of our specific LSTM. Within the setting we tested—recurrent PointNav agents with five matched-protocol sensor/gaze configurations—the takeaway is that sensor structure and gaze location are tunable axes along which the *format* of learned spatial memory can be steered; both are invisible under the uniform-vision assumption of prior work. Open questions include whether richer gaze architectures (stochastic gaze with entropy bonus, information-seeking intrinsic reward, attention-based token selection) produce active gaze and how that reshapes representational content; whether the same five-condition ablation on a transformer backbone reproduces or refines our findings; whether the effects persist at longer navigation horizons and with other sensor topologies; and whether analogous representational-content effects arise in non-navigation tasks or non-visual modalities.

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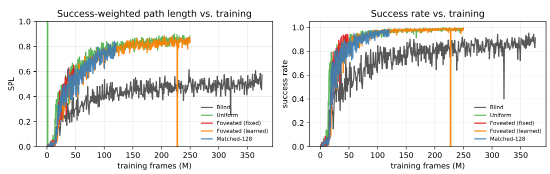
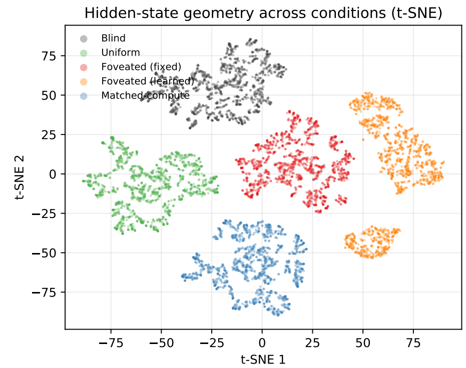
A Supplementary visualisations



(a) Training curves (TensorBoard event-file scalars). All four sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon. The glitch at frame $\approx$ 228M (fov-learned) is a resume-boundary artefact in the scalar log.



(b) t-SNE projection of top-layer LSTM hidden states pooled across all five conditions ($1{,}500$ random steps each; perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§4.3).

Figure 8: Supplementary visualisations referenced from the main text.

B Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation In the first foveated-learned training run the gaze decoder collapsed (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \to 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$ where $g \in [0,1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma^2_{\text{target}} = 0.01$ (std ≈ 0.1), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

Pilot 1 (uncapped, job 2840256) After $10\,$M frames, gaze variance exceeded that of a uniform distribution over $[0,1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at $\sim 8\,$M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288) Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

C Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈170M frames of stable progress (reward ≈ 10, SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL$=0$ until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.